Bhavin K. Moriya



Data Scientist | Machine Learning | Privacy-Preserving Analytics
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LinkedIn | GitHub | Google Scholar

Profile

Data scientist with a **Ph.D. in Mathematics** and 2+ years building production data-science tools in German applied research (BMBF project, Hochschule Esslingen). Rare combination of **statistical modelling, machine learning, and homomorphic-encryption engineering** (OpenFHE/C++, TFHE-rs/Rust). 17 peer-reviewed publications. Seeking data science / quantitative modelling roles in industry.

Experience

Data Scientist / Research Assistant — **Hochschule Esslingen, Germany** — **Nov 2023 – Dec 2025**

- **AnoMoB (BMBF research project)**: Engineered first-of-its-kind encrypted comparison operations and non-linear function evaluation (square roots, polynomials) using **OpenFHE (C++)** and **TFHE-rs (Rust)** for privacy-preserving mobility analytics.
- **Benchmarking study:** Co-authored a performance comparison of runtime and memory consumption of the **CKKS and TFHE** schemes; published on the Cryptology ePrint Archive (2025).
- **Data engineering:** Built production-quality tools for statistical analysis of large mobility datasets using **Python, polars, and geopandas**.
- **Mobility simulation:** Simulated traffic with **SUMO** and compared travel modalities by CO_2 emissions and route duration using OpenRouteService/OpenTripPlanner (Dockerized architecture).
- **Decision-support tool:** Developed a weighted-scoring application (Streamlit) that lets users trade off travel time against CO_2 across transport modes.

Assistant Professor — **Universidade Federal de Viçosa, Brazil** — **Nov 2015 – Nov 2023**

- Taught undergraduate and master's-level mathematics; supervised 3 master's dissertations.
- Led a FAPEMIG-funded international research project; published 16+ peer-reviewed articles.

Postdoctoral Fellowships: IMSc Chennai (2011–2013) | HRI Allahabad (2013–2014) | CMI (2014) | University of Brasília (2014–2015)

Skills

Programming: Python (expert), C++, Rust, SQL, Go, Julia
Data & ML: polars, pandas, geopandas, scikit-learn, PyTorch, Streamlit, Docker, Git
Specialised: Homomorphic encryption (OpenFHE, TFHE-rs), Monte Carlo simulation, regression & classification, time-series forecasting
Domain: Credit-risk modelling (PD/LGD/CCF), derivative pricing (Black–Scholes–Merton), portfolio optimisation (CAPM, Markowitz)
Languages: English (fluent), German (A2, actively working towards B1), Portuguese (fluent), Hindi & Gujarati (native)

Education

- **Ph.D. in Mathematics**, Harish Chandra Research Institute — 2011
  Thesis: Some Zero Sum Problems in Combinatorial Number Theory
- **M.Sc. & B.Sc. in Mathematics**, The M.S. University of Baroda — 2003 – 2005

Selected Projects

- **Credit-Risk Model (PD · LGD · CCF)** — end-to-end probability-of-default, loss-given-default, and credit-conversion-factor modelling pipeline.
- **FEM Solver** — finite-element-method solver written from scratch in Python.
- **AudioMNIST** — spoken-digit recognition system (neural networks, DSP).
- **Audio DSP Web App** — interactive tool exploring the mathematics of audio digital signal processing. Demo
- **Vascular Surrogate Models** — machine-learning surrogates accelerating intracranial-aneurysm CFD.

Publications & Awards

- (2025) **A Performance Comparison of the Homomorphic Encryption Schemes CKKS and TFHE**, Cryptology ePrint Archive.
- 17 peer-reviewed publications in mathematics and cryptography; full list: Google Scholar.
- International grants and fellowships: ICM (Seoul 2014), CIMPA–ICTP, NBHM, FAPEMIG.